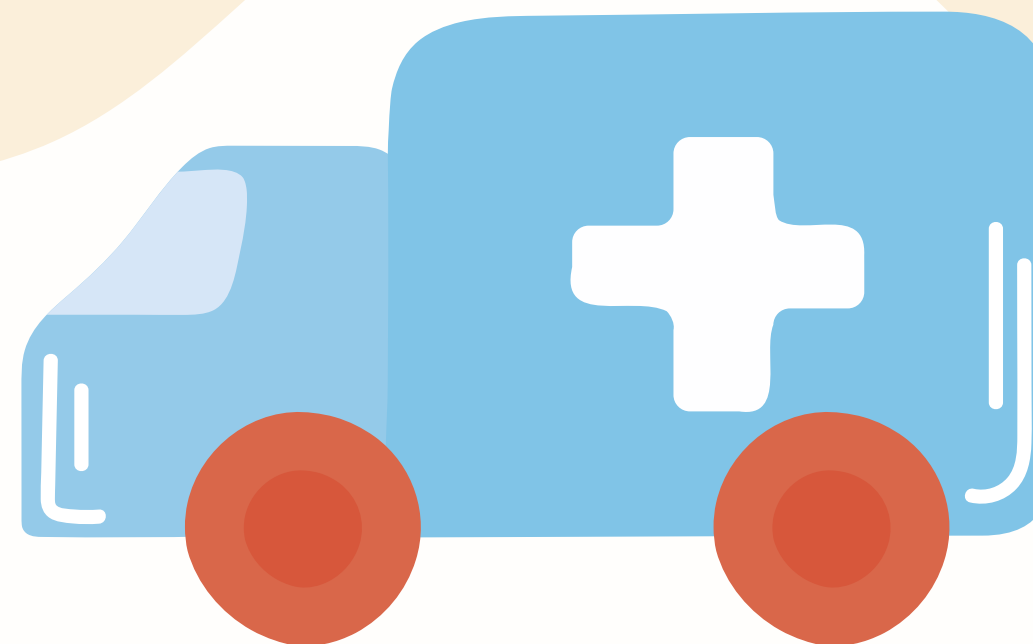
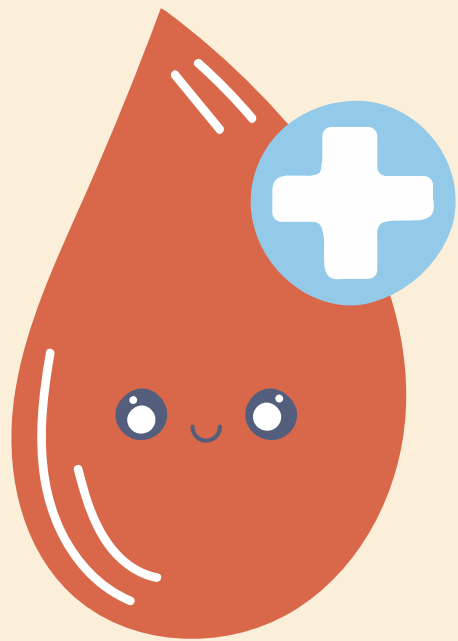
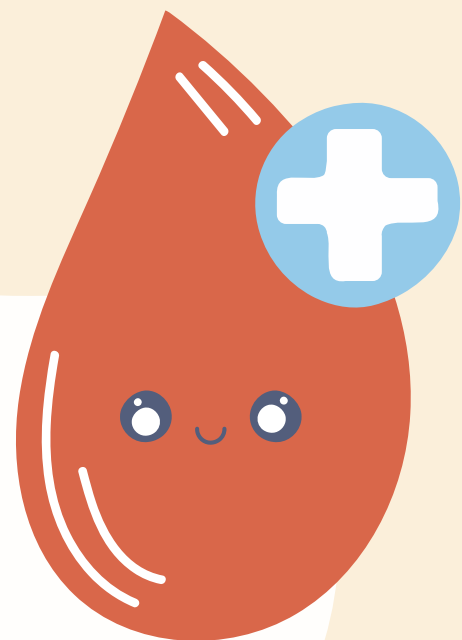
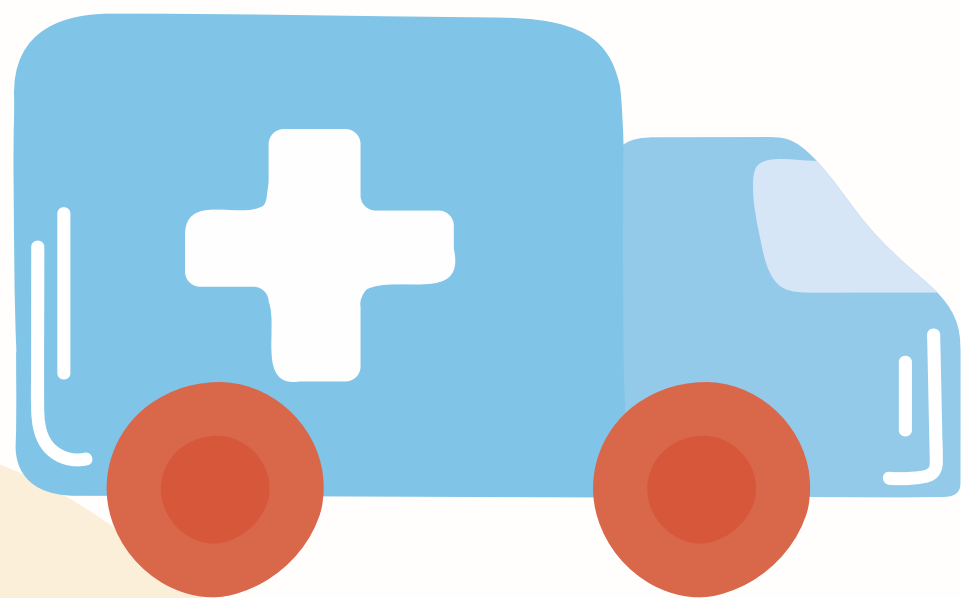


TRIAGE AI

Education about hygiene,
vaccines, and mental health
strengthens communities and
saves lives.





TriageAI is an end-to-end AI-powered emergency response platform that transforms every stage of the mass

casualty pipeline from field triage to hospital arrival.

At the field level, paramedics describe injuries via voice, text, or photo and Google Gemini AI instantly classifies severity (Red/Yellow/Green/Black), recommends medical actions, and flags surgical or blood transfusion needs.

At the dispatch level, the system integrates with India's 108/112 ambulance network to match patients with the nearest hospital that has the right specialty, available beds, ICU capacity, and operating rooms using live data synced from the e-Hospital/ABDM portal.

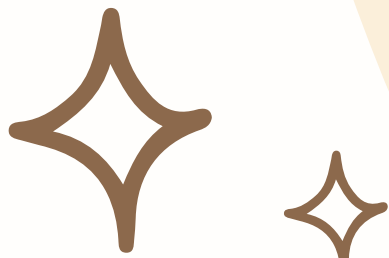
At the hospital level, ER teams receive pre-arrival alerts with the AI assessment, injury type, and surgical needs so they can prepare before the ambulance arrives.

The real-time command dashboard gives incident commanders full visibility across patients, resources, ambulances, and hospital capacity.



OPPORTUNITIES

a. How different is it from any of the other existing ideas?



Current Landscape	TriageAI Improvement
START/SALT triage done manually with paper tags	AI classifies severity in 10 seconds via voice/text/photo
BHISHM cubes have AI for inventory, NOT triage decisions	First tool using multimodal AI for triage classification
108 ambulance drivers call hospitals manually to check beds	Live bed availability synced from e-Hospital + AI hospital matching
Hospitals get no data about incoming patients	Pre-arrival alerts with AI assessment sent before ambulance arrives
Ayushman Bharat (ABDM) digitizes records, not triage	TriageAI feeds INTO ABDM real-time patient data from field to hospital



TECHNOLOGIES TO BE USED IN THE SOLUTION

Technology	Category	Purpose
Google Gemini API (gemini-2.0-flash)	Google AI	Multimodal triage classification: text, vision, medical reasoning
Google Maps Platform	Google Service	Hospital routing, distance/ETA, turn-by-turn ambulance navigation
React 19 + Vite 8	Frontend	Modern component-based web application
Web Speech API	Browser API	Real-time voice-to-text for voice triage mode
Vercel	Cloud Hosting	Production deployment with global CDN, auto-HTTPS
108/112 ERSS API	India Emergency	Ambulance dispatch integration (designed for)
e-Hospital / ABDM	India Health Stack	Live hospital bed availability data sync (designed for)
Lucide React	UI Components	Premium icon system
CSS Custom Properties	Design System	Complete dark-mode design system with triage color coding

B. HOW WILL IT BE ABLE TO solve the problem?

- . Reduces triage classification time from 2 minutes to 10 seconds per patient
- . Eliminates human bias and fatigue related errors in severity classification
- . Prevents hospital overloading by distributing patients based on real time capacity
- . Cuts "golden hour" delays by 40–60% through optimized routing
 - Enables hospital ER teams to prepare in advance with pre-arrival data

C. USP OF THE PROPOSED SOLUTION

ONLY MULTIMODAL AI TRIAGE TOOL IN INDIA VOICE, PHOTO, AND TEXT IN ONE PLATFORM
END-TO-END PIPELINE NOT JUST TRIAGE, BUT DISPATCH + ROUTING + HOSPITAL COORDINATION
ZERO INFRASTRUCTURE WORKS ON ANY SMARTPHONE BROWSER,
NO APP INSTALLATION
DESIGNED FOR INDIAN EMERGENCY ECOSYSTEM
INTEGRATES
WITH 108/112, E-HOSPITAL, ABDM, BHISHM

LIST OF FEATURES OFFERED BY THE SOLUTION

FIELD TRIAGE MODULE:

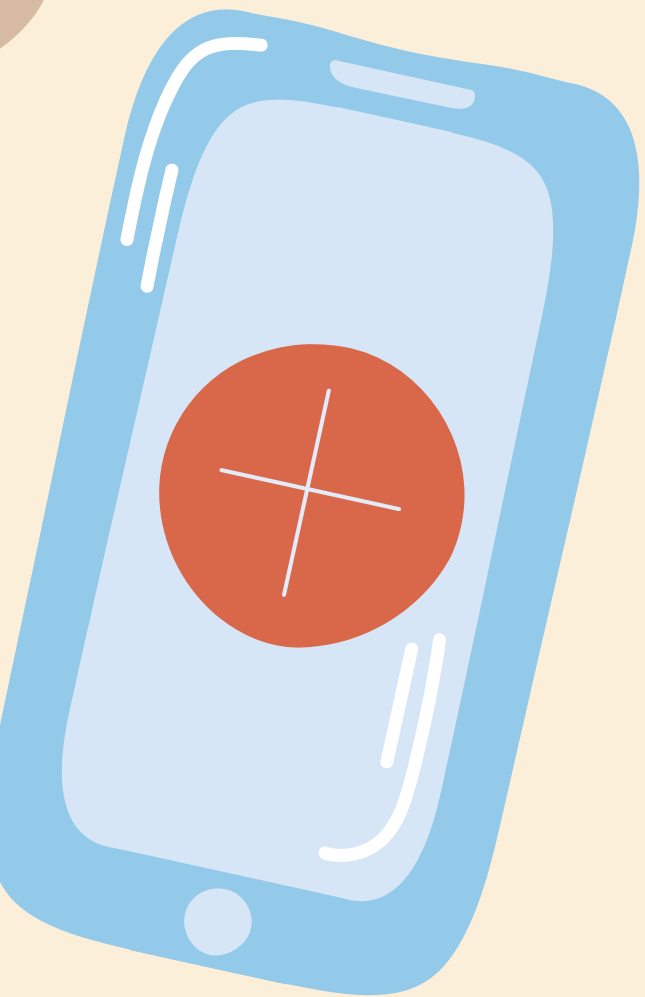
VOICE TRIAGE: SPEAK INJURIES, GEMINI TRANSCRIBES AND CLASSIFIES SEVERITY IN REAL TIME

PHOTO ASSESSMENT: UPLOAD/CAPTURE INJURY PHOTOS, GEMINI VISION ANALYZES VISIBLE INJURIES

TEXT TRIAGE: TYPE DESCRIPTIONS FOR AI CLASSIFICATION

4-LEVEL SEVERITY CLASSIFICATION: RED/YELLOW/GREEN/BLACK FOLLOWING INTERNATIONAL START PROTOCOL

AI MEDICAL ASSESSMENT: DETAILED ANALYSIS WITH CONFIDENCE SCORE, SURGERY/BLOOD FLAGS, RECOMMENDED ACTIONS



AMBULANCE & DISPATCH MODULE:

108/112 Integration - Connected to India's Emergency Response Support System

Smart Hospital Matching: AI matches injury type - hospital specialty + distance + live bed count

Pre-Arrival Alerts: Hospital ER receives patient data + AI assessment before ambulance arrives

Live Bed Availability: Real-time bed/ICU/ventilator/OT data synced from e-Hospital portal

Turn-by-Turn Routing: Google Maps-powered optimal route for ambulance drivers



Command Center Module:

Real-Time Dashboard: Live crisis event tracking with patient counts, severity distribution

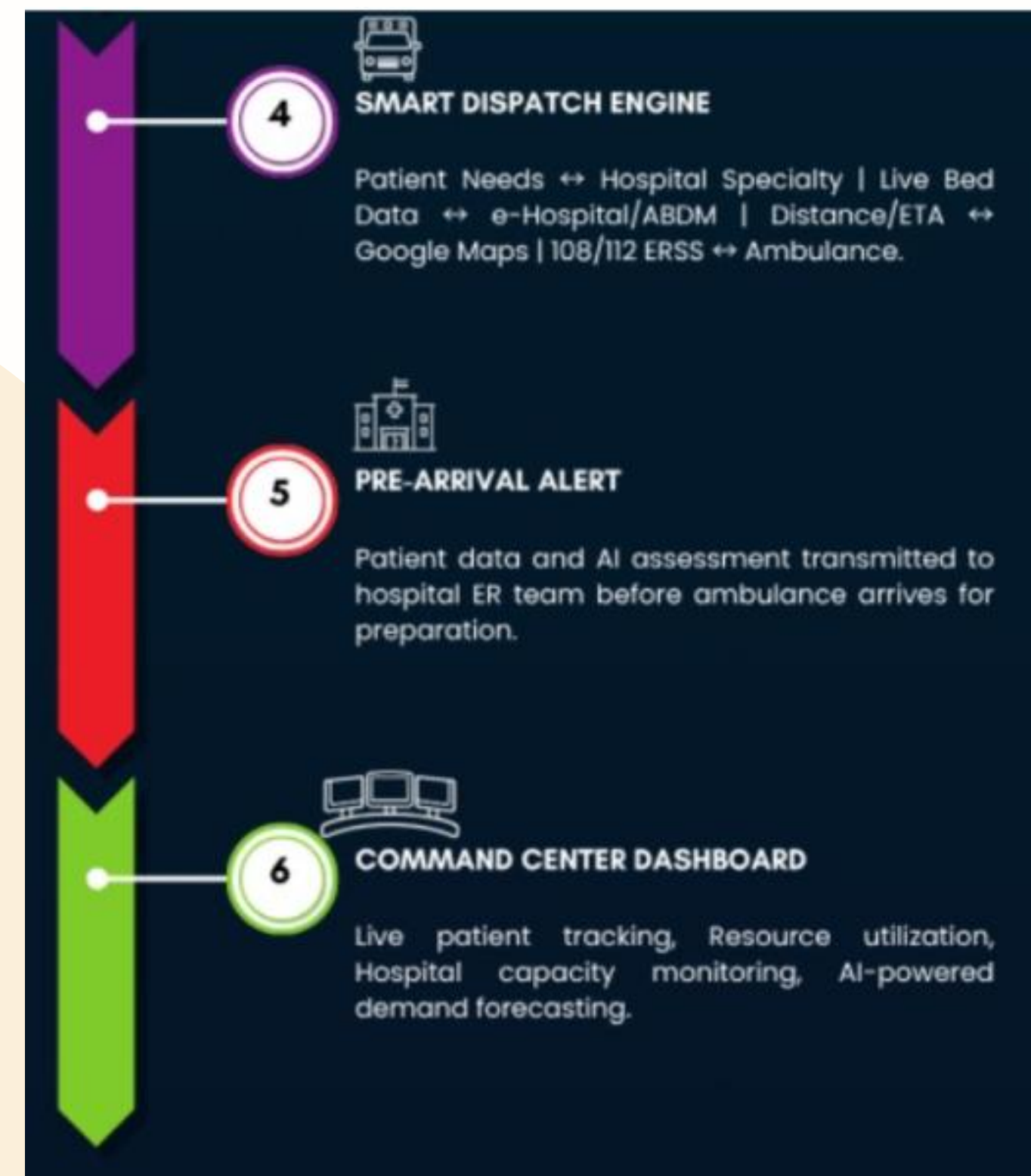
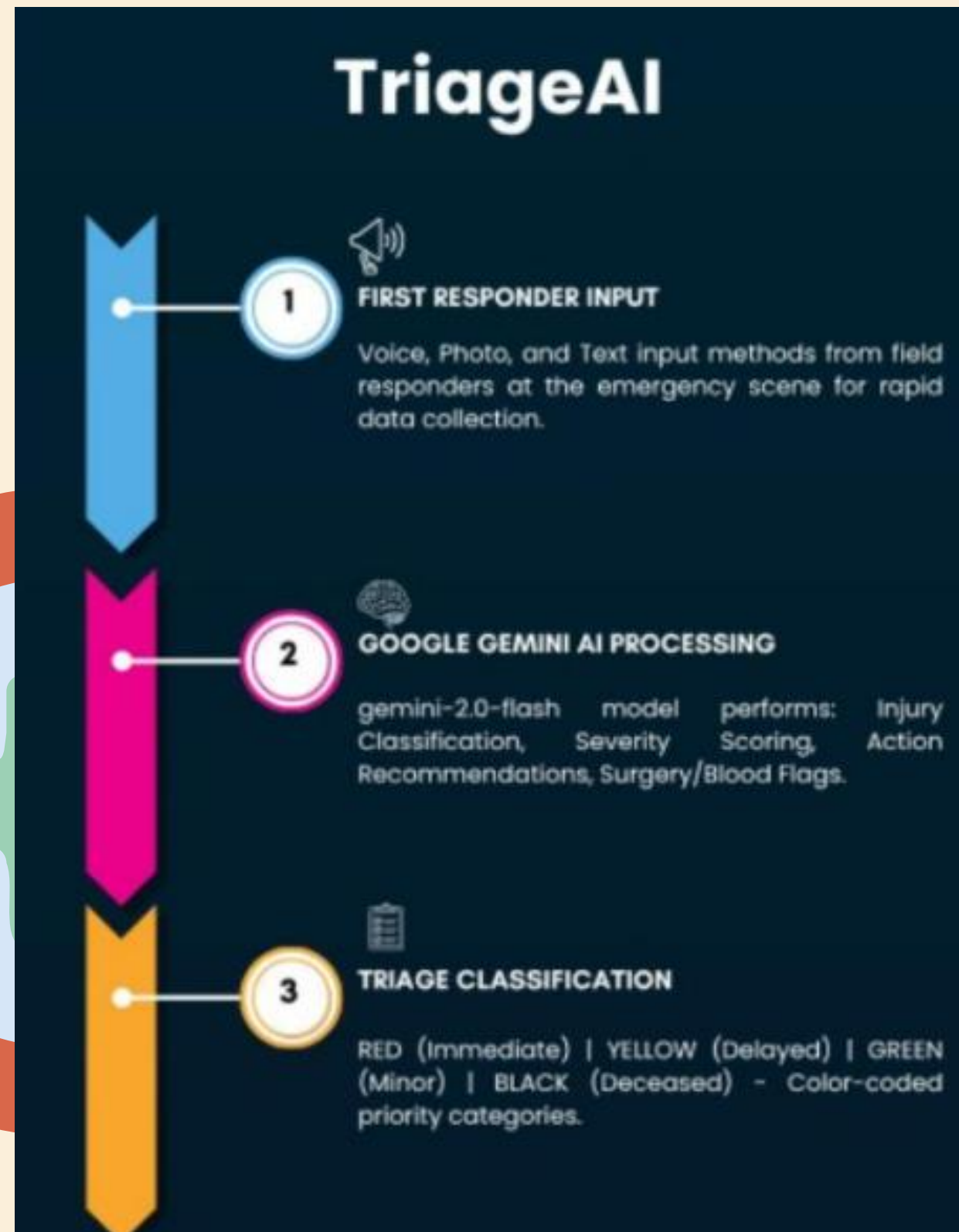
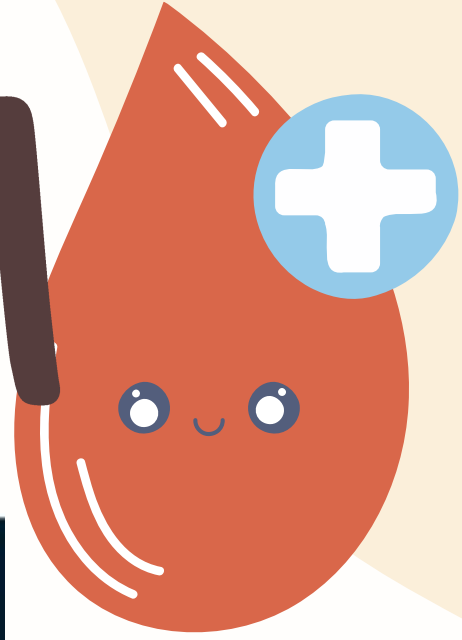
Resource Management: Track ambulances, beds, surgeons, blood supply, ventilators, operating rooms

Patient Registry: Searchable, sortable registry with full triage history

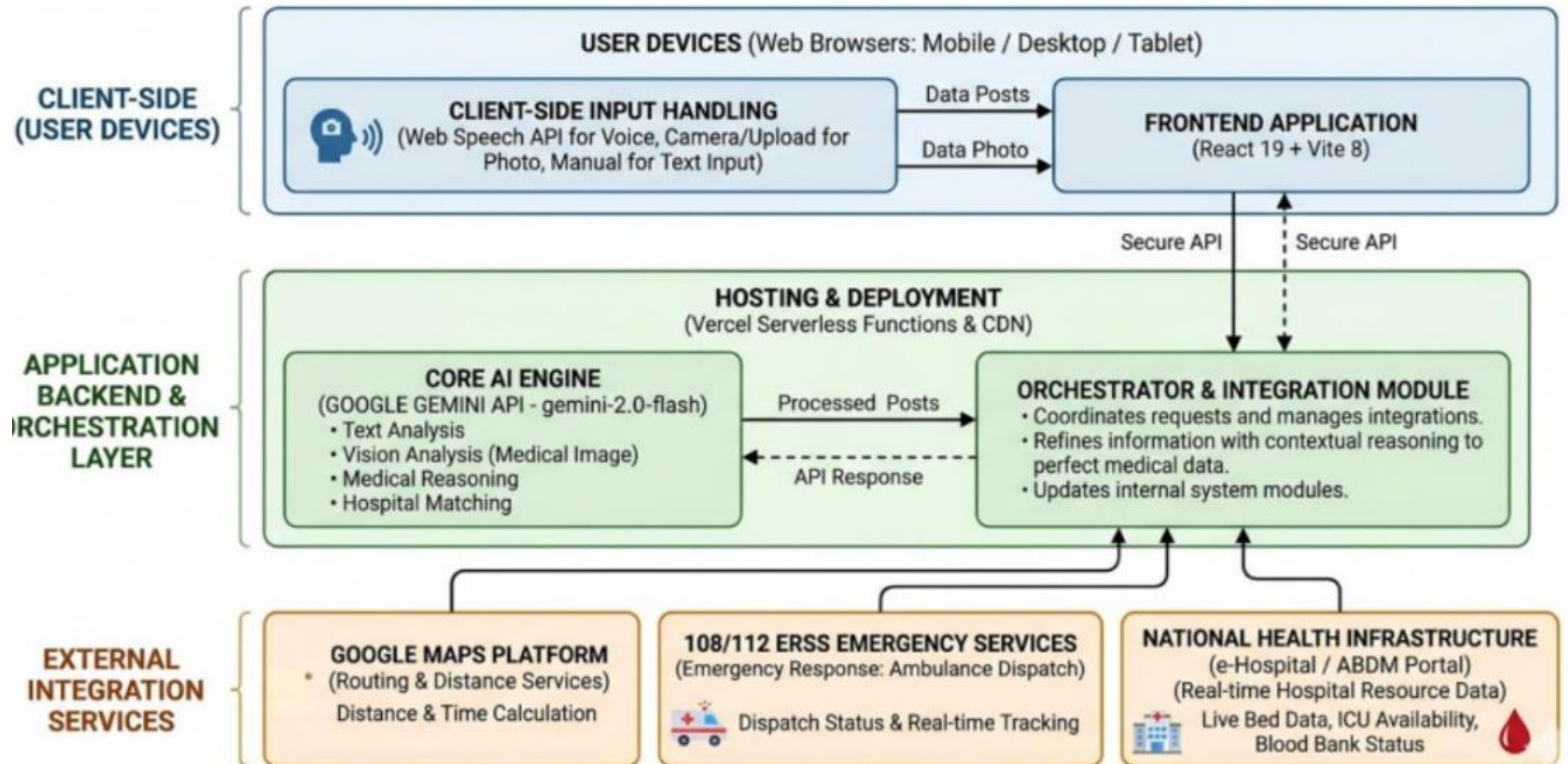
Hospital Network Map: Visual overview with capacity and routing info

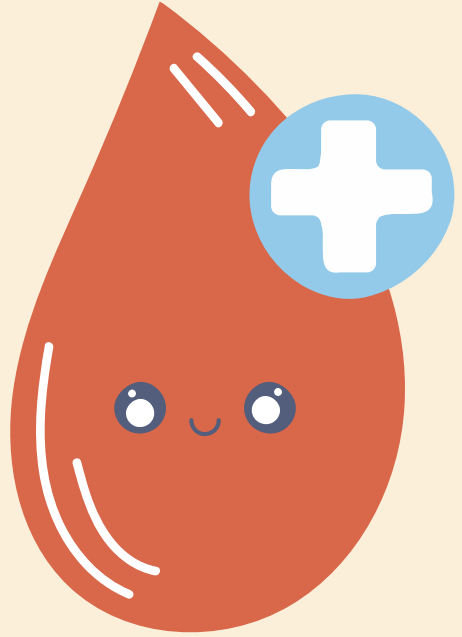


PROCESS FLOW DIAGRAM



ARCHITECTURE DIAGRAM OF THE PROPOSED SOLUTION



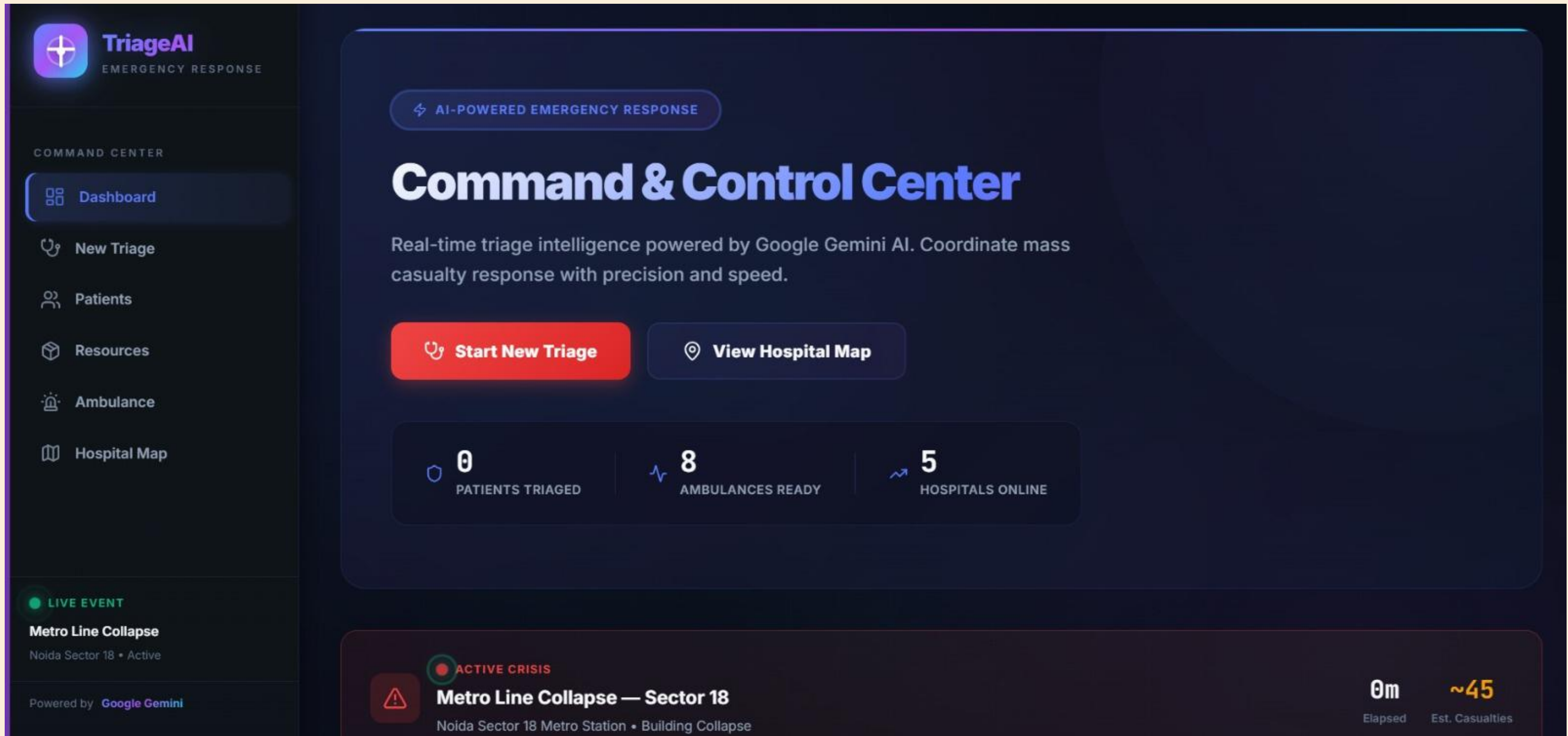


SCALABILITY:

Stateless architecture scales horizontally via Google Cloud
Designed for India's health infrastructure (ABDM, e-Hospital, NHA)
Can support 10,000+ concurrent users across multiple disaster sites



Triage AI



TriageAI

EMERGENCY RESPONSE

COMMAND CENTER

Dashboard

New Triage

Patients

Resources

Ambulance

Hospital Map

LIVE EVENT

Metro Line Collapse

Noida Sector 18 • Active

Powered by Google Gemini

Text

Voice

Photo

NAME

AGE

GENDER

Alice

19

Female

INJURY DESCRIPTION

Nose Bleeding, Semi-conscious

Analyze with Gemini AI

AI Assessment

IMMEDIATE — Life Threatening

Confidence: 65% • Priority Score: 9/10

MEDICAL ASSESSMENT

Patient presents with described symptoms. Classified as Immediate priority based on keyword analysis. Full AI analysis requires API key configuration.

INJURY TYPE

Mixed trauma

RECOMMENDED SPECIALTY

Trauma Surgery

SURGERY REQUIRED

BLOOD TRANSFUSION

RECOMMENDED ACTIONS

1. Assess airway, breathing, circulation

2. Monitor vital signs

3. Prepare for transport

COLORED SEVERITY CARDS

TriageAI

EMERGENCY RESPONSE

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Dashboard

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LIVE EVENT

Metro Line Collapse

Noida Sector 18 • Active

Powered by Google Gemini

AI Resource Forecast

Based on 0 triaged patients (0 critical, 0 delayed), estimated resource demand: 0 bed-hours, 0 surgical interventions, 0 ambulance dispatches.

Ambulances

Emergency medical transport vehicles

812

AvailableTotal Capacity

812

Hospital Beds

Available beds across all nearby hospitals

3250

AvailableTotal Capacity

3250

Surgeons on Duty

Trauma and emergency surgeons available

58

AvailableTotal Capacity

58

Blood Units

Available blood units in blood bank

145200

AvailableTotal Capacity

145200

Ventilators

Mechanical ventilation support units

1115

AvailableTotal Capacity

1115

Operating Rooms

Available surgical operating theaters

46

AvailableTotal Capacity

46

QUICK DATA COLLECTION



Emergency Network Integration

Connected to 112 ERSS (Emergency Response Support System) and 108 Ambulance Network. Live bed availability synced from e-Hospital portal. GPS routing powered by Google Maps Platform.

7

Active Units

3

Available

Pending Dispatches

Patients awaiting ambulance assignment — sorted by severity



No pending dispatches

All triaged patients have been dispatched or are minor severity

Active Dispatch

Ambulance driver view — turn-by-turn routing



Select a patient to dispatch

Choose a patient and hospital from the left panel

LIVE
UPDATES

SAVETIME
SAVE LIFE

TriageAI
EMERGENCY RESPONSE

COMMAND CENTER

Dashboard

New Triage

Patients

Resources

Ambulance

Hospital Map

AIIMS Trauma

Level 1 Trauma

120
Beds

2.3
km
Distance

8 min
ETA

0
Patients

Get Directions

Safdarjung Hospital

Emergency Medicine

85
Beds

3.1
km
Distance

12 min
ETA

0
Patients

Get Directions

Apollo Emergency

Multi-Specialty

60
Beds

5.7
km
Distance

18 min
ETA

0
Patients

Get Directions

Max Super Speciality

Cardiac & Neuro

45
Beds

7.2
km
Distance

22 min
ETA

0
Patients

Get Directions

Noida Sector 18, Delhi NCR

Showing 5 hospitals in 10km radius

Powered by Google Maps Platform



LIVE EVENT

Metro Line Collapse

Noida Sector 18 • Active

THANK YOU SO MUCH!

Check our App

Our App

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Contributors: *404 Brain Not Found*

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